Introduction

AI & Gen AI

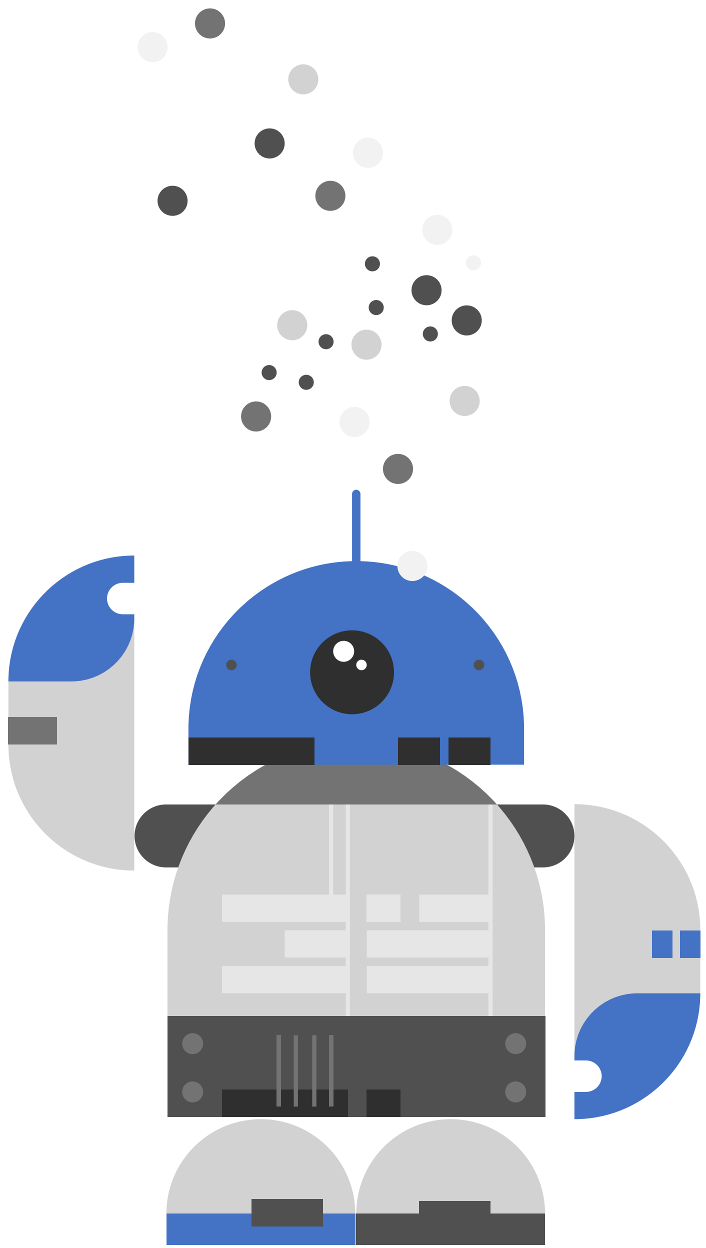
What is the most complex problem you have solved with AI?

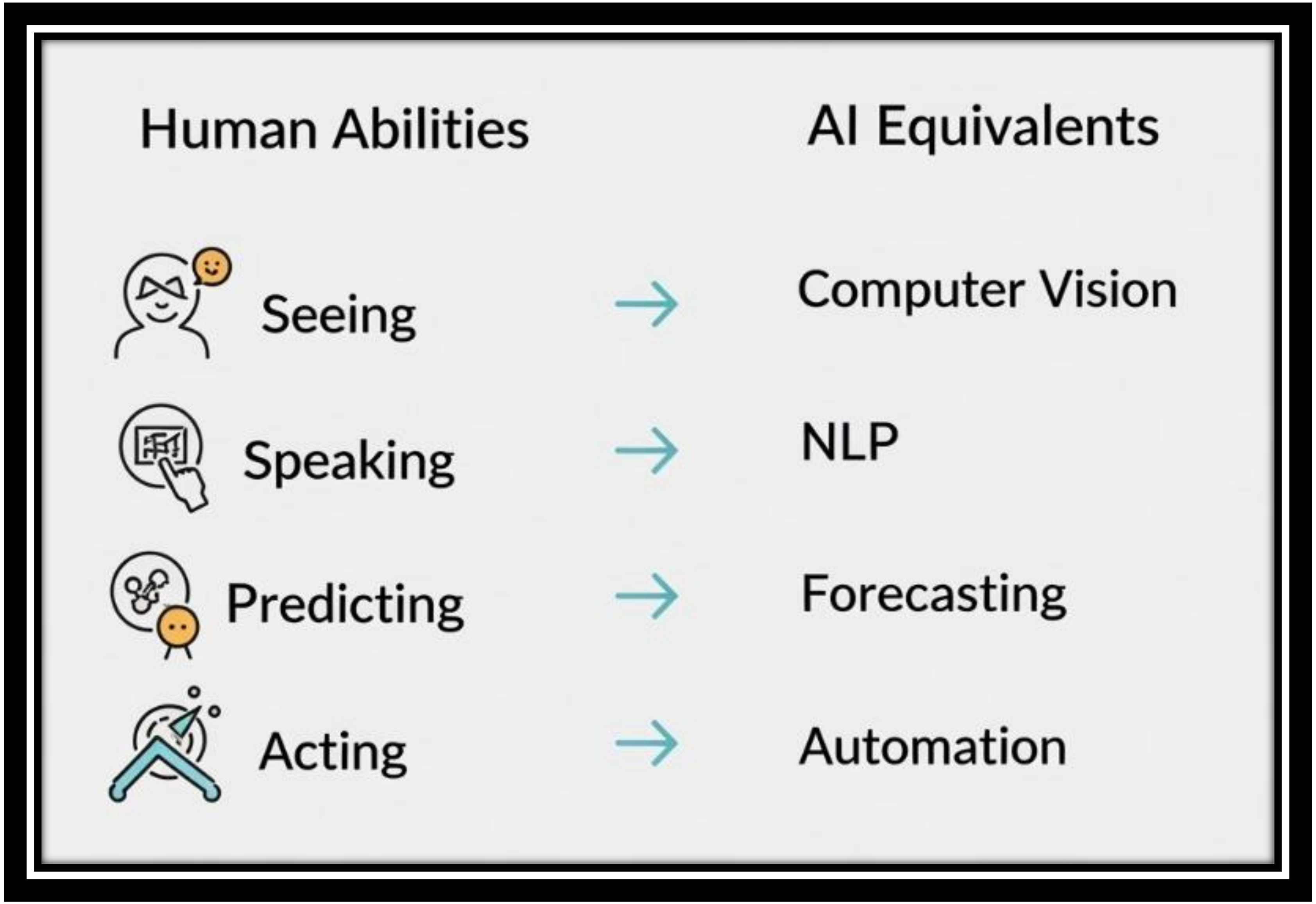
What is So Special About Working With AI?

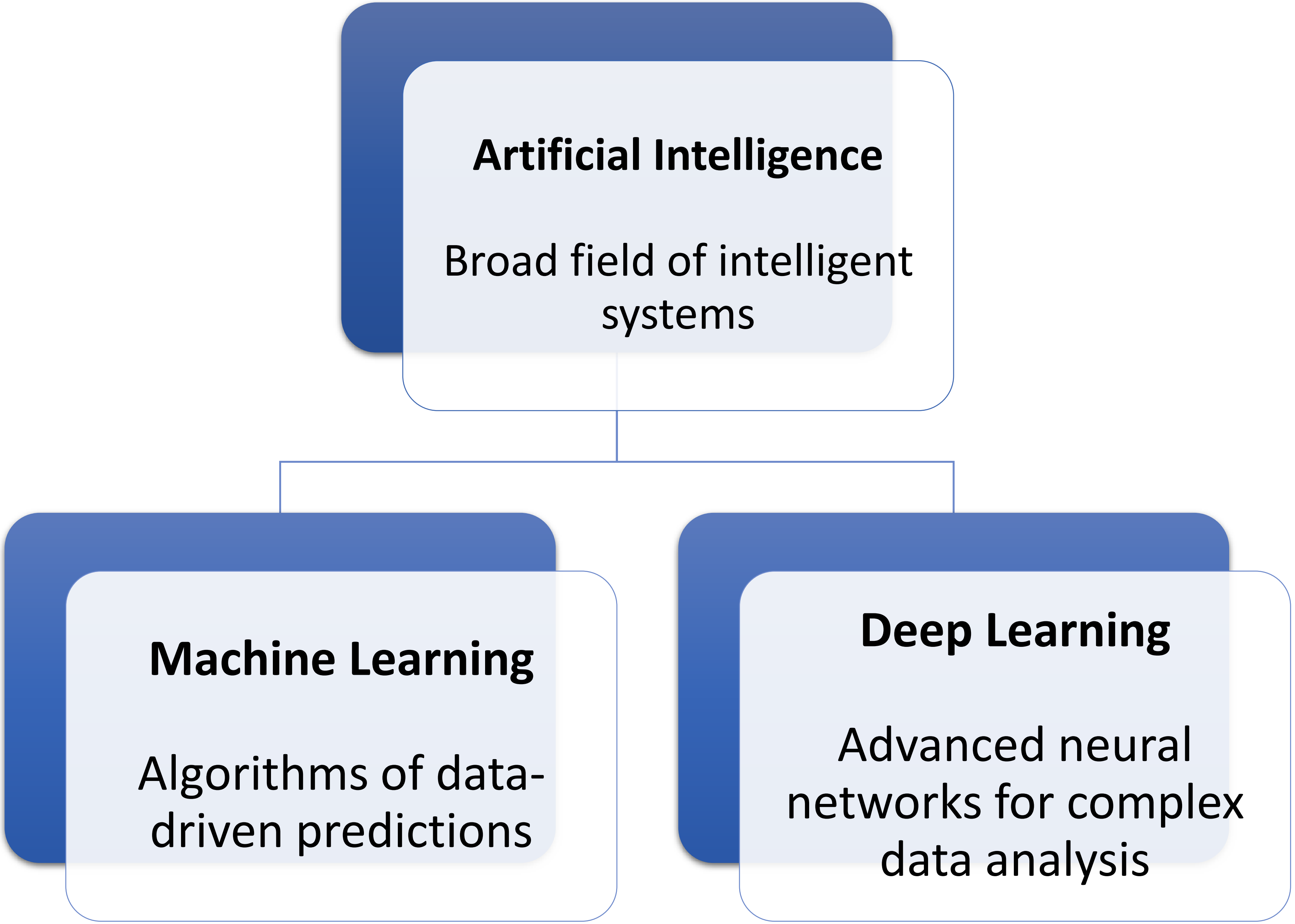
AI is a Journey of Pursuit of Intelligence

Artificial Intelligence Defined

A set of algorithms designed to perform tasks that typically require human intelligence. Used in multiple areas like making decisions, recognizing patterns, pictures and speeches.




Human Abilities
AI Equivalents
Seeing
Computer Vision
Speaking
NLP
Predicting
Forecasting
Acting
Automation


Artificial Intelligence
Broad field of intelligent systems
Machine Learning
Algorithms of data-driven predictions
Deep Learning
Advanced neural networks for complex data analysis

Can Machines Think?

1. **1950: The Question**
   Alan Turing's paper first posed this revolutionary question.
2. **1955: Term Coined**
   John McCarthy introduced "artificial intelligence" to secure workshop funding.
3. **Global Impact**
   The concept sparked imagination and investment despite technological limitations.
4. **Turing Test**
   AI defined as systems exhibiting behavior interpreted as human intelligence.

The General Problem Solver

In 1956, Newell and Simon created the General Problem Solver. It aimed to solve problems as mathematical formulas.

1

Mathematical Formulas

Represent problems as mathematical formulas.

2

Key Component

Newell and Simon's physical symbol system hypothesis.

3

General Intelligence

Symbols were key to general intelligence.

The Limits of Computer Understanding

Computers don't understand the purpose of the game. They flex their ability to follow rules. Computer and human intelligence act differently.

1. **Rules**
   Computers follow rules and pattern matching.

2. **No Awareness**
   They are unaware of the concept of a game.

Machine Learning Breakthrough

1959

First Self-Learning Program

Arthur Samuel's checkers program that could beat its programmer

∞

Pattern Recognition

ML creates expert lists and matches patterns automatically

M+

Data-Driven Learning

Learning from millions of conversations and examples

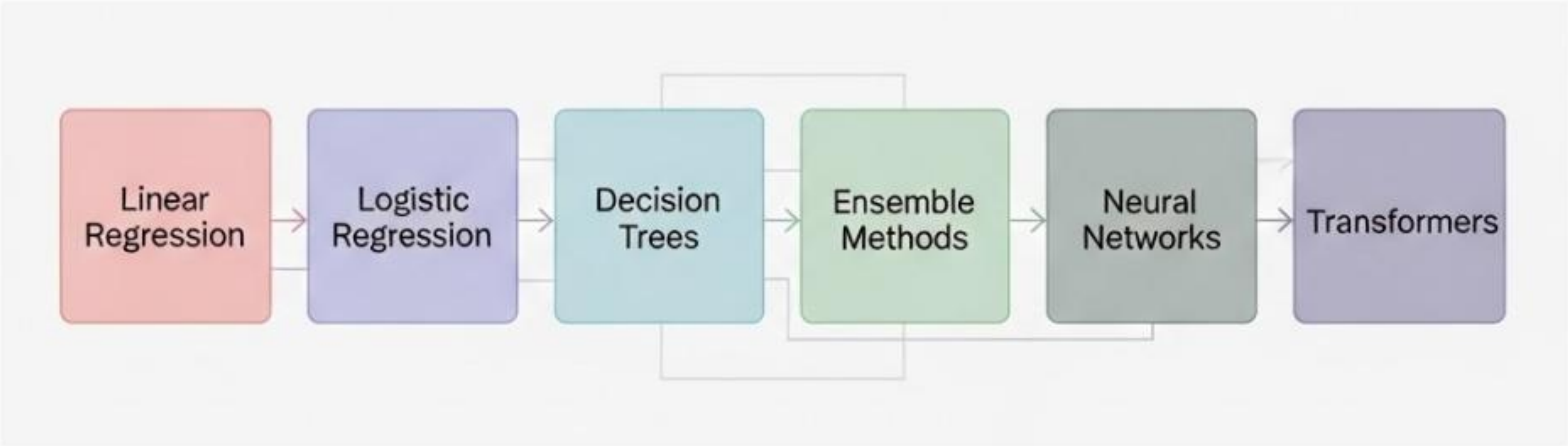
Machine learning emerged when traditional "if-then" statements became too complex for tasks like animal identification. ML can identify patterns without knowing what to look for.

Machine learning is a branch of artificial intelligence that focuses on developing systems that can learn from and make decisions based on data. Rather than being explicitly programmed to perform specific tasks, these systems identify patterns in data and improve their performance over time.

Algorithm which detects patterns from larger data sets which are used to make predictions. It learns relationships between variables and labels

Company	Objective / Goal	Machine Learning Use	Approach / Methods
NETFLIX	To maximize user engagement for retention	Leverages collective behavior to make accurate predictions	Uses user attributes for personalized recommendations
STARBUCKS™	Maximizing revenue by influencing ordering preferences	Uses mobile app and loyalty card to collect data	Suggests new products based on customers' ordering preferences

Behind every AI application lies an algorithm, which can be thought of as a decision-making engine. Some commonly used ones are:



Linear and Logistic Models

- Identify trends and calculate probabilities. For example, forecasting monthly sales or predicting the chance of customer churn.

Decision Trees

- Breakdown choices step by step, like a flowchart for decision-making. They are widely used in fraud detection and customer eligibility checks.

Ensemble Methods (Random Forest, XGBoost)

- Combine multiple decision trees to improve accuracy, applied in fraud detection and risk scoring.

Neural Networks

- Modeled loosely on the brain, these handle complex inputs such as images or speech.

Transformers

- The foundation of modern AI systems like ChatGPT, enabling breakthroughs in language understanding and generation. These algorithms are not abstract theories—they are embedded in products and services already shaping industries.

Steps for Creating a Model

Collecting and preprocessing data

- Ensure data is clean and ready for analysis by handling missing values, normalising data and splitting it into training and testing
- The training sample is used to learn patterns, and the testing sample is used to predict outcomes

Identifying relevant features

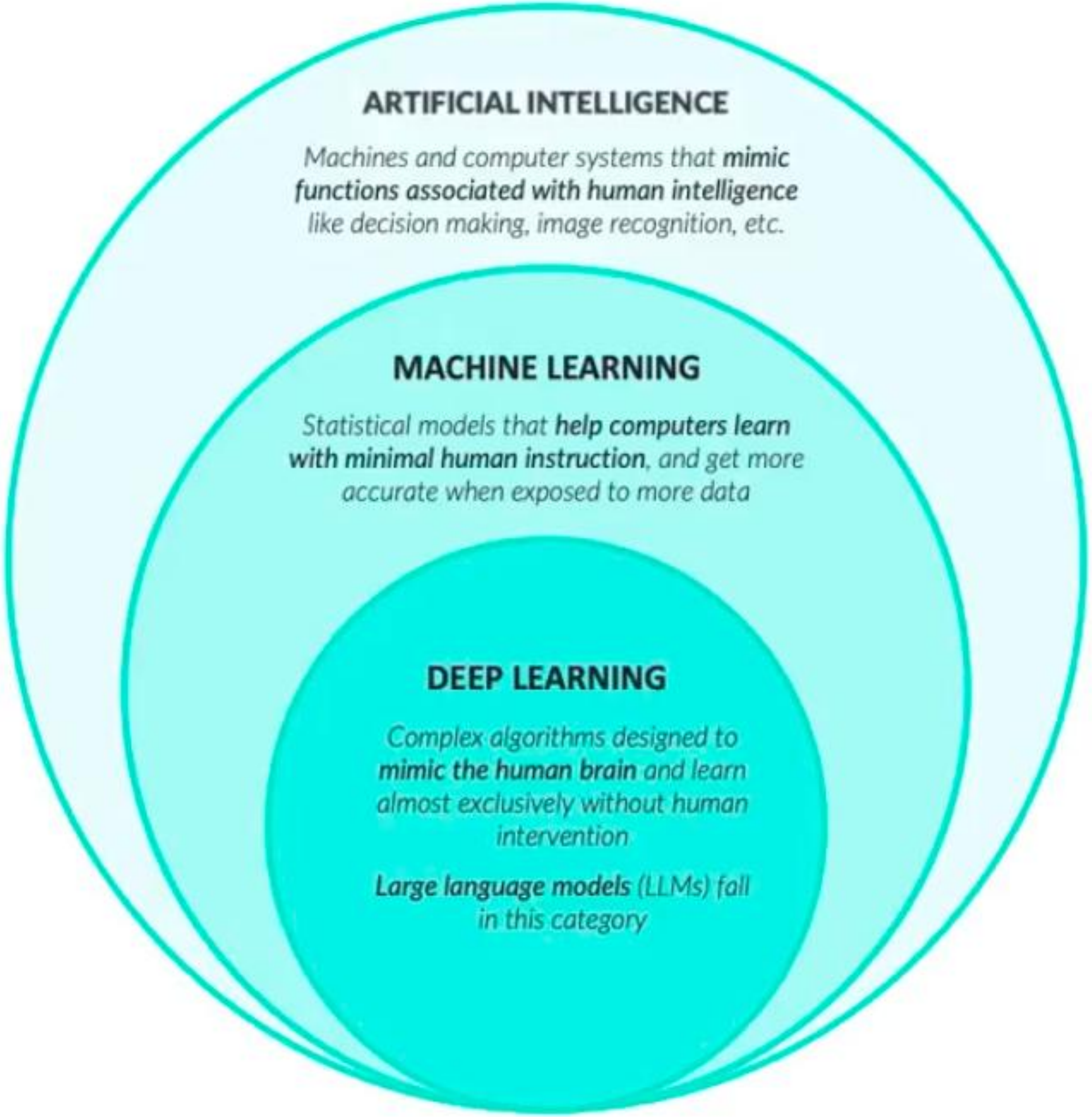
Techniques that assess which features are important predictors help determine the features with the greatest impact on outcomes.

Building the model

- The training dataset is used to build machine learning models like decision trees, logistic regressions, or neural networks. These models learn patterns from the data.

Reasons for AI Failures?

- **Chasing buzzwords:** Investing in AI because it is fashionable, not because it solves a real business problem.
- **Scope creep**: Starting too big instead of beginning with focused, achievable use cases.
- **Data readiness issues**: Underestimating the importance of clean, reliable data. Without strong data foundations, even the most advanced algorithms produce weak outcomes.



"Weak" vs. "Strong" Artificial Intelligence

Weak AI *(current state)*

- Can only perform **specialized tasks**
- For example, GPT-4 could tell you how to drive a car, but couldn't use that knowledge to drive a car itself

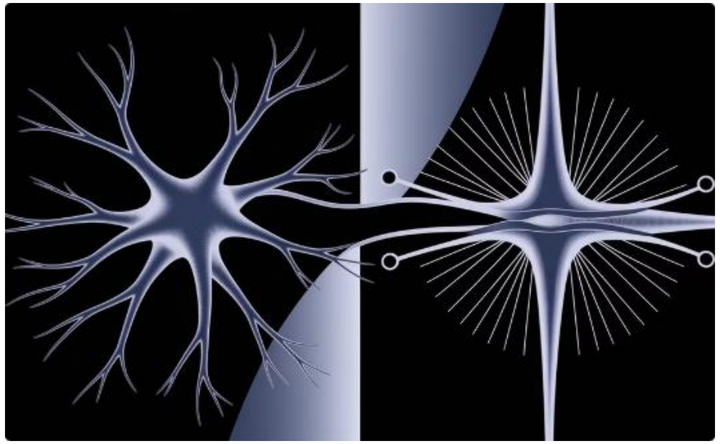
Strong AI *(Artificial General Intelligence)*

- Can learn and perform **any task** that a human can do
- Its development is still on the horizon

While Deep Learning has its foundations in Statistics, it focuses solely on **producing accurate model outputs** (not understanding the underlying data or relationships)

These models are often called "black boxes" since they can pick up nuances in the data that humans can't understand or detect

Artificial Neural Networks



Brain-Inspired Design

ANNs mimic the structure of the human brain with billions of connected neurons.

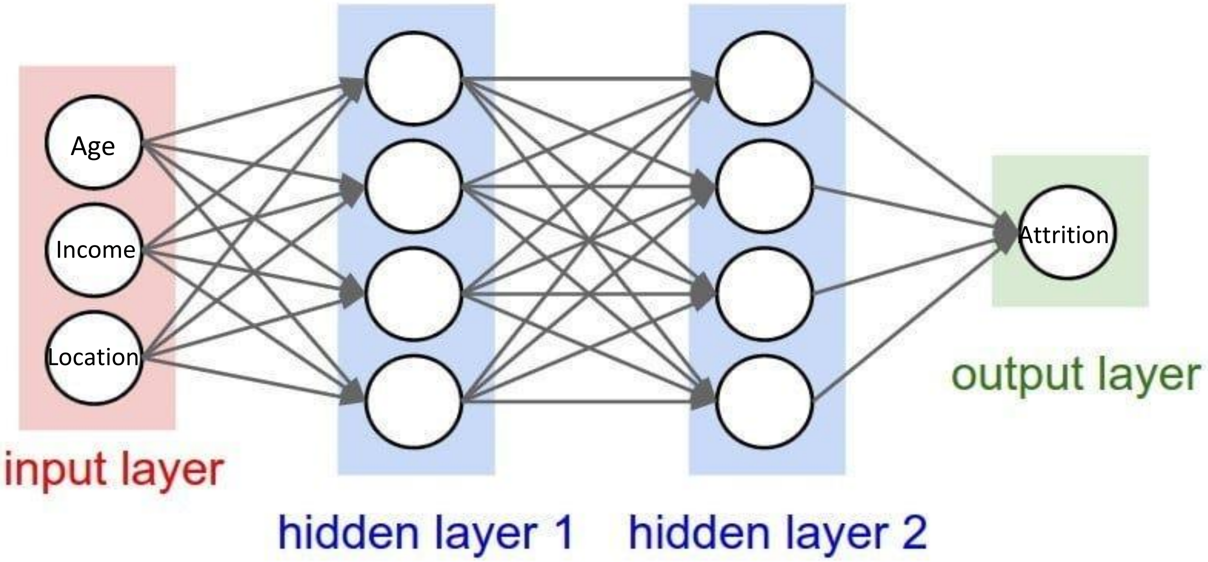
Strengthening Connections

Neurons increase connection strength based on experiences, improving performance over time.

Coordinated Learning

Like a marching band, neural networks coordinate complex patterns through simple rules.

Artificial Neural Networks



Key Elements of the Neural Network

Input Layer: This is where data enters the system. In this example, customer information (age, income, location, purchase history) is fed into the network.

Hidden Layers: These middle layers are where the "thinking" happens. The network discovers patterns and relationships in the data that might not be obvious to humans.

Output Layer: This produces the final result or prediction. In this example, it's predicting whether a customer will make a purchase.

How Neural Network Works

- Think of each circle (or "neuron") as a decision-making unit.
- The connections between neurons (the lines) represent how information flows and how important each piece of information is.
- When the network processes new information:
    1. Data enters through the input layer
    2. Each hidden layer performs calculations to identify patterns
    3. The output layer provides the final prediction or classification

- Initial layers detect simple edges and shapes
- Deeper layers identify more complex structures (faces or even objects)

Hierarchical learning process enables deep learning modules to understand and interpret data with remarkable accuracy.

How Tesla Leverages AI?

- Tesla uses deep learning simulations to test drivers
- Users provide valuable data to help Tesla improve its software
- Simulations are key to the Full Self-Driving Beta program
- High safety standards for deployment
- The program tests and refines autonomous driving with real user feedback
- AI-powered autonomous driving creates opportunities for test drivers to refine the technology
- This iterative process ensures the system becomes safer and more reliable over time

AI: Influence in Organisations

Cognitive

Routine, cognitive

For example: office assistants, sales agents, brokers and drivers.

Non- routine, cognitive

For example: engineers, management, healthcare workers and designers.

Manual

Routine, manual

For example: factory workers, construction workers and mechanics.

Non- routine, manual

For example: nurses, waiters and security staff.

Routine

Non-routine

Applying AI to Your Organization

Think about tasks computers do well. Does your work have set rules and possibilities?

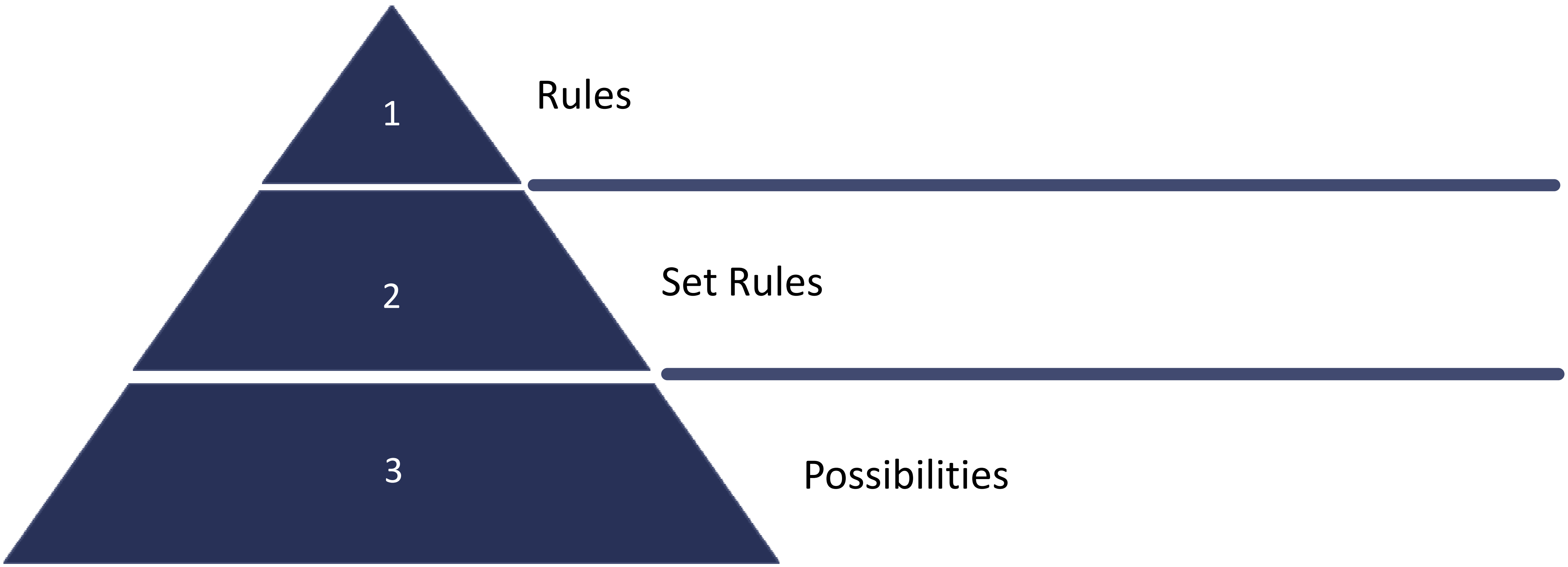
Pattern Matching

Set Rules

Possibilities

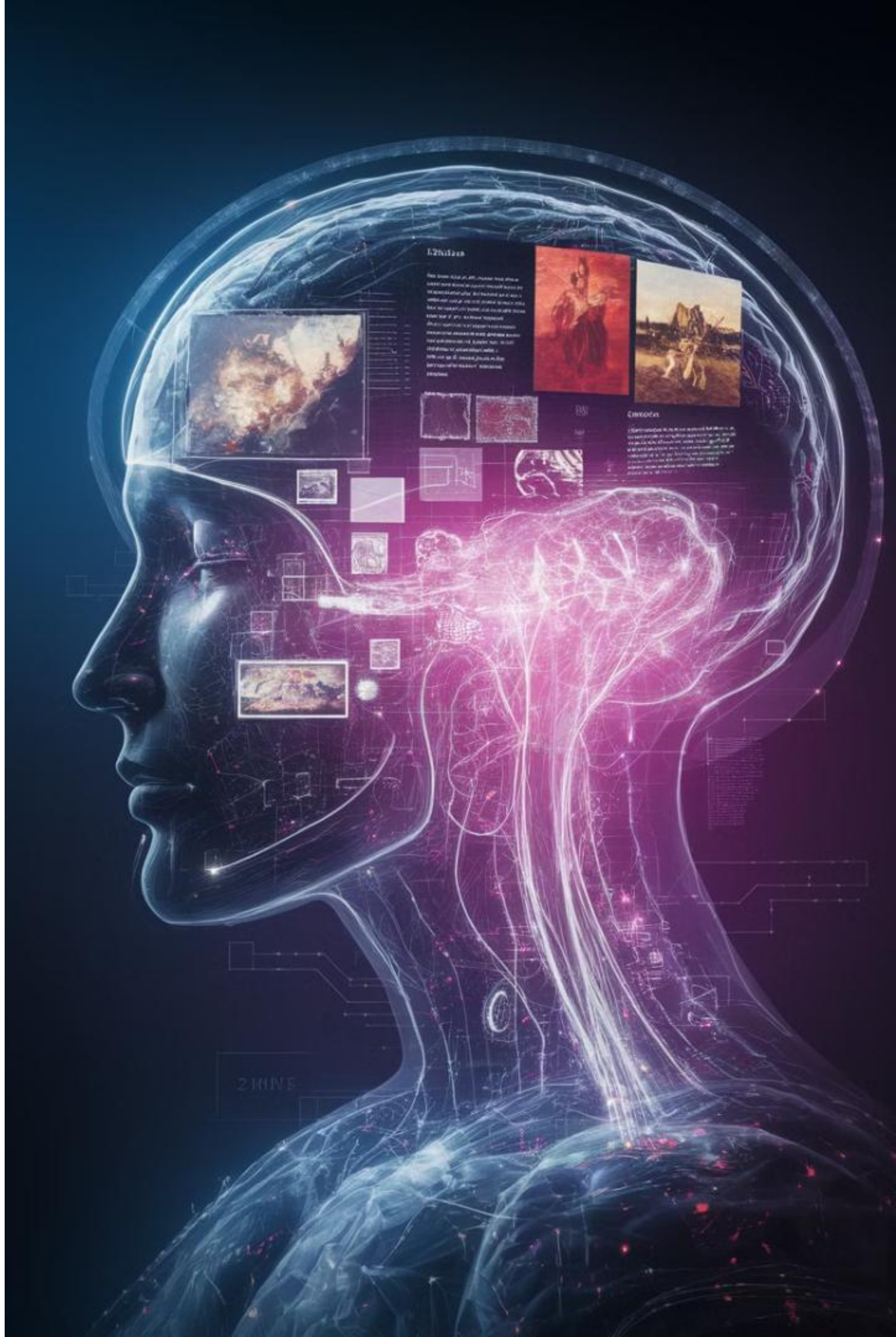
First to Benefit from AI

AI will be first to benefit within a space with set rules. Focus on work that benefits from artificial intelligence.



Introduction to Generative AI

AI systems that autonomously create new content like text, images, audio, and video



GENERATIVE PRE-TRAINED TRANSFORMERS

Generative Pre-Trained Transformers (GPTs) are a type of large language model trained on massive text datasets, and are designed to generate outputs that mimic human-written text

GENERATIVE

The model generates **new and original natural language text**, instead of copying and pasting existing data

PRE-TRAINED

The model **was already trained on a large dataset** before being fine-tuned to perform specific tasks

TRANSFORMERS

A type of deep learning model that can **process sequential inputs** and differentiate the importance of individual parts *(also known as self-attention)*

LLMs like ChatGPT are among the **most sophisticated deep learning models** ever built; GPT-4, which powers premium versions of ChatGPT, has over 1 TRILLION parameters, cost over $100 million dollars, and took 11 months to train

GENERATIVE AI & LLMs

Generative AI systems are deep learning models capable of generating original text, images and other types of media in response to user prompts

- **Large Language Models** (LLMs) are generative AI models focused on producing text outputs specifically
- Other generative AI models include DALL-E and Midjourney, which are used to generate images

A large language model:

When did Paris become the capital of France?

Paris became the capital of France in the 6^{th} century.

How does the model answer the question?

1. It looks at millions of documents for similar questions and related statements like ***"in 508 A.D., Paris became the capital of France"***
2. It is then associates the prompt with these documents, and does its best to **mimic the language** from the responses
3. Since it has an element of randomness to account for its uncertainty, if you ask it again it will likely tell you something **similar but not identical**

Large language models like ChatGPT are **COLOSSAL achievements in machine learning** that model the "shape" of language

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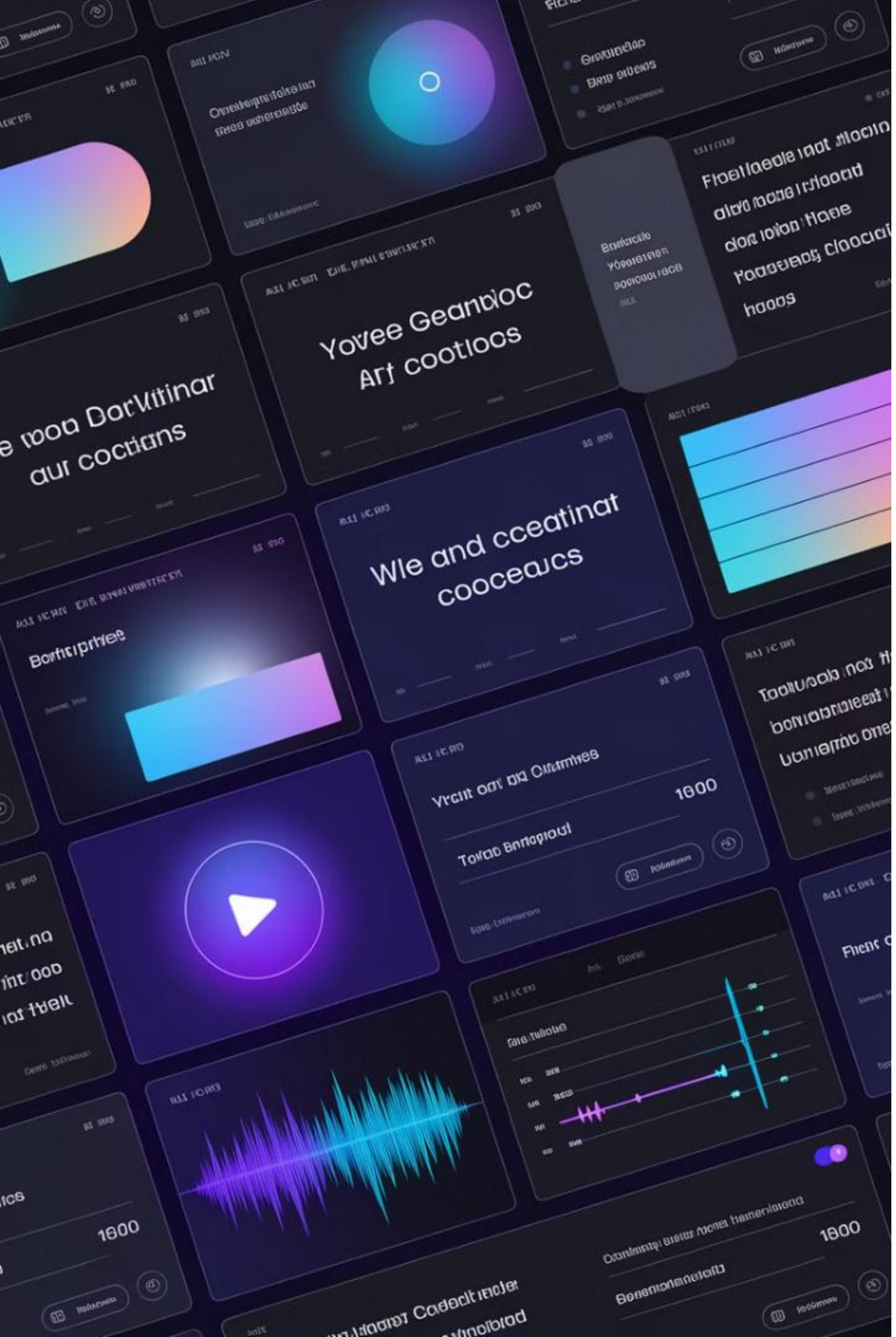
A "simple" language model:

How does the model fill in the blank?

- The model doesn't *know* that "Paris" is the correct response, but suggests **the answer it thinks is most probable** in the given context
- Put simply, it compared ***"Paris"*** with thousands of words and determined that it had the strongest relationship with words like ***"capital"*** and ***"France"***
- This may seem easy, but remember that the word *"capital"* has multiple meanings, that France has had other capitals, and that the blank could also be things like ***"beautiful"***, ***"a popular tourist destination"***, etc.



Creates Content

Text, images, videos, and sounds

Versatile Applications

Analysis, classification, optimization, recommendations

Recent Advances

Transformers, diffusion models, massive computing power

Key Technical Architectures

Transformers

Neural networks using self-attention to identify essential input elements

Diffusion Models

Transform random noise into target images through gradual steps

RLHF

Uses human feedback signals to improve model performance

WARNING: COMMON PITFALLS

LLMs are known to **"hallucinate" facts** with total confidence

- *Remember that YOU are ultimately responsible for verifying the accuracy of model outputs*

Solutions provided **may be suboptimal or entirely incorrect**

- *AI tools don't guarantee accuracy, and may provide incorrect or inefficient solutions*

These tools are broad and often **lack specific domain knowledge**

- *LLMs may not understand specific business context or the "why" behind the responses they produce*

LLMs are NOT capable of **common sense** or **human judgement**

- *Models need specific, objective inputs, and may miss critical context that may seem obvious to humans*

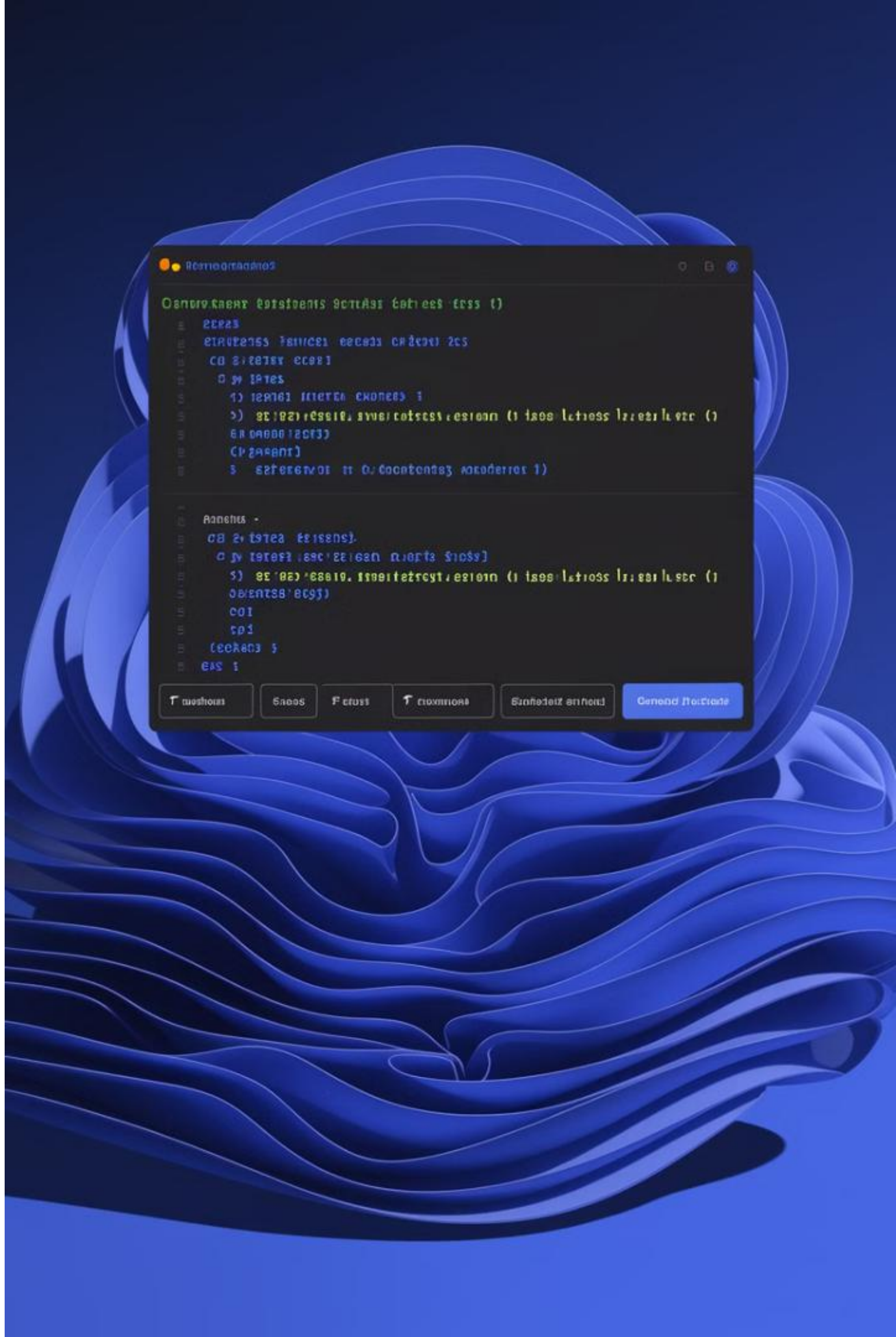
Gen AI Key Terms

1. **Prompt**
   Instructions given to AI

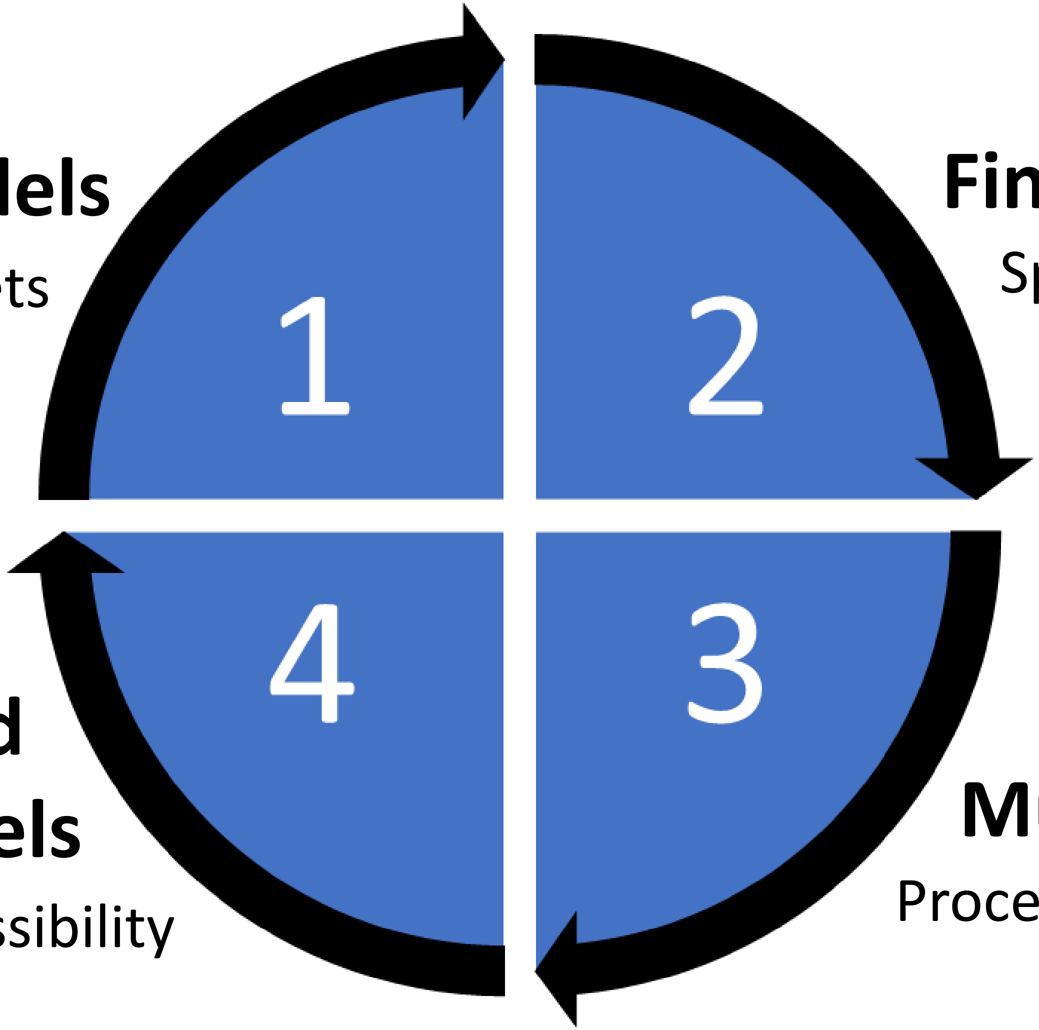
2. **Token**
   Unit of text processing

3. **Temperature**
   Controls randomness of output

4. **Hallucination**
   Incorrect or fabricated information



Overview of AI Models



Foundation Models

Trained on massive datasets

Fine-tuned Models

Specialized for specific tasks

Multimodal Models

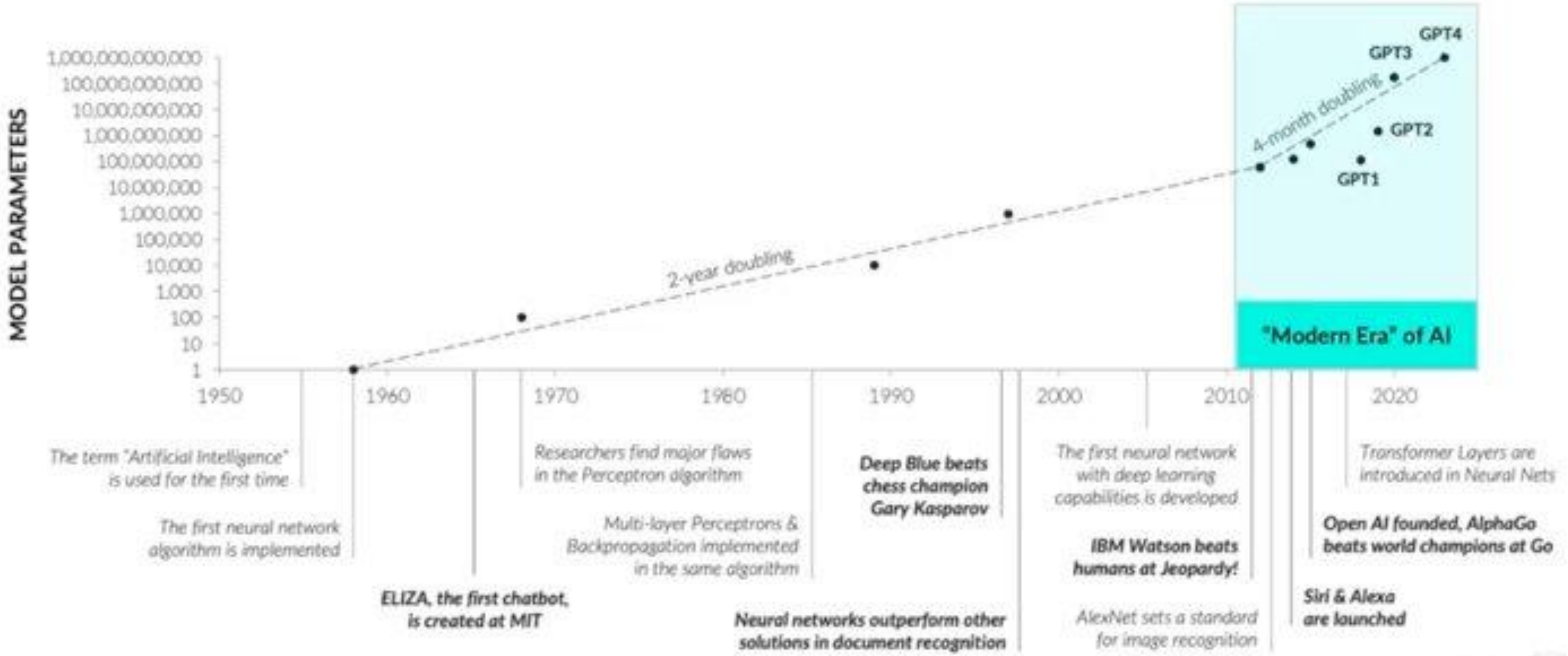
Process multiple input types

Open vs. Closed Models

Varying degrees of accessibility

A BRIEF HISTORY OF AI

AI tools like **ChatGPT** became widely popular in late 2022, but owe their success to more than **60 years of research and development** in artificial intelligence systems



How Do We Measure AI?

Measuring AI Effectiveness

Accuracy, Precision & Recall: How well does the system make predictions. Critical in applications where errors are costly such as fraud detection or healthcare.

Latency: How quickly does it respond. Critical for customer facing apps

Throughput: Can AI model handle large volumes of data or requests.

Total cost of ownership(TCO): What does it cost to build, run and maintain.

Time to Value: How fast benefits be realized after implementation

The Enterprise AI Stack Overview